

Removing Limits from Calculus: A Discrete Approach

Caylee Spivey (UConn)

Mathematics Continued Conference

March 8th, 2026

We want to develop a process for using calculus on sequences that mirrors the traditional calculus used on functions.

We will use two main tools to do this: the difference operator and falling powers.

Motivating Examples

Let's start by comparing “nice” formulas for finite sums with their integral analogues:

Sum of First n Numbers

We have

$$\sum_{k=0}^n k = 1 + 2 + \cdots + n = \frac{n(n+1)}{2} = \frac{n^2}{2} + \frac{n}{2}$$

while

$$\int_0^n x \, dx = \frac{n^2}{2}.$$

Motivating Examples

Sum of First n Squares

$$\sum_{k=0}^n k^2 = \frac{n(n+1)(2n+1)}{6} = \frac{n^3}{3} + \frac{n^2}{2} + \frac{n}{6}$$

while

$$\int_0^n x^2 \, dx = \frac{n^3}{3}.$$

Motivating Examples

Sum of First n Squares

$$\sum_{k=0}^n k^2 = \frac{n(n+1)(2n+1)}{6} = \frac{n^3}{3} + \frac{n^2}{2} + \frac{n}{6}$$

while

$$\int_0^n x^2 \, dx = \frac{n^3}{3}.$$

Sum of Powers of 3

$$\sum_{k=0}^n 3^k = 1 + 3 + 9 + \cdots + 3^n = \frac{3^{n+1} - 1}{2}$$

while

$$\int_0^n 3^x \, dx = \frac{3^n - 1}{\ln 3}.$$

The Difference Operator

Recall

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}.$$

For $h \in \mathbb{Z}$, the least nonzero choices are ± 1 .

The Difference Operator

Recall

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}.$$

For $h \in \mathbb{Z}$, the least nonzero choices are ± 1 .

Definition

The function $(\Delta f)(n) = f(n+1) - f(n)$, with $h = 1$, is called the **discrete difference** of f and Δ is called the **difference operator**. Explicitly,

$$(\Delta f)(0) = f(1) - f(0)$$

$$(\Delta f)(1) = f(2) - f(1)$$

$$(\Delta f)(2) = f(3) - f(2)$$

$$\vdots$$

Applying the Difference Operator

In the following examples, take $n = 0, 1, 2, \dots$

$$f(n) = n^2$$

$f(n)$	0	1	4	9	16	25	...
$(\Delta f)(n)$	1	3	5	7	9	...	

In general,

$$\begin{aligned}(\Delta f)(n) &= (n+1)^2 - n^2 \\&= n^2 + 2n + 1 - n^2 \\&= 2n + 1.\end{aligned}$$

Compare to $(x^2)' = 2x$.

Applying the Difference Operator

Example: $f(n) = n^3$

$$\begin{aligned}(\Delta f)(n) &= (n+1)^3 - n^3 \\ &= 3n^2 + 3n + 1\end{aligned}$$

Compare to $(x^3)' = 3x^2$.

Applying the Difference Operator

Example: $f(n) = n^3$

$$\begin{aligned}(\Delta f)(n) &= (n+1)^3 - n^3 \\ &= 3n^2 + 3n + 1\end{aligned}$$

Compare to $(x^3)' = 3x^2$.

Example: $f(n) = 3^n$

$f(n)$	1	3	9	27	81	243	...
$(\Delta f)(n)$		2	6	18	54	162	...

$$(\Delta f)(n) = 3^{n+1} - 3^n = (3-1)3^n = 2 \cdot 3^n = 2f(n)$$

So $\Delta f = 2f$.

Compare to $g(x) = 3^x \implies g'(x) = \ln 3 \cdot 3^x = \ln 3 \cdot g(x)$

Properties of Δf

Properties

- ① Δ is linear: $\Delta(f + g) = \Delta f + \Delta g$ and $\Delta(Cf) = C\Delta f$.
- ② $\Delta f = \Delta g$ if and only if $f = g + C$ for some constant C .
- ③ $\Delta f = 0$ if and only if f is constant.
- ④ f is a polynomial sequence of degree d ($d \geq 1$) if and only if Δf is a polynomial sequence of degree $d - 1$.
- ⑤ $\Delta f = af$ if and only if $f(n) = C(a + 1)^n$ with $C = f(0)$.

Properties of Δf

Properties

- ① Δ is linear: $\Delta(f + g) = \Delta f + \Delta g$ and $\Delta(Cf) = C\Delta f$.
- ② $\Delta f = \Delta g$ if and only if $f = g + C$ for some constant C .
- ③ $\Delta f = 0$ if and only if f is constant.
- ④ f is a polynomial sequence of degree d ($d \geq 1$) if and only if Δf is a polynomial sequence of degree $d - 1$.
- ⑤ $\Delta f = af$ if and only if $f(n) = C(a + 1)^n$ with $C = f(0)$.

For the last property, compare to $g' = a \cdot g \iff g(x) = Ce^{ax}$ with $C = g(0)$.

So $f(n) = 2^n$ with $a = 1$ is a discrete analogue of e^x since $\Delta f = f$.

Theorem

For a sequence $\{f(n)\}_{n \geq 0}$, $\Delta f = 0 \iff \{f(n)\}$ is constant.

Vanishing of Δ

Theorem

For a sequence $\{f(n)\}_{n \geq 0}$, $\Delta f = 0 \iff \{f(n)\}$ is constant.

Proof.

($\Leftarrow$) Obvious.

($\Rightarrow$)

$$\begin{aligned}(\Delta f)(n) = 0 &\iff f(n+1) - f(n) = 0 \text{ for all } n \\ &\iff f(n+1) = f(n) \text{ for all } n.\end{aligned}$$

So $f(0) = f(1) = f(2) = \dots$. Thus, $\{f(n)\}$ is constant. □

Fundamental Theorem of Discrete Calculus

Theorem (FTDC)

Let $\{f(n)\}$ be a function on $n \geq 0$. If $\{F(n)\}$ is any sequence such that $(\Delta F)(n) = f(n)$ for all $n \geq 0$, then

$$f(0) + f(1) + \cdots + f(n) = F(n+1) - F(0) \text{ for all } n \geq 0.$$

Fundamental Theorem of Discrete Calculus

Theorem (FTDC)

Let $\{f(n)\}$ be a function on $n \geq 0$. If $\{F(n)\}$ is any sequence such that $(\Delta F)(n) = f(n)$ for all $n \geq 0$, then

$$f(0) + f(1) + \cdots + f(n) = F(n+1) - F(0) \text{ for all } n \geq 0.$$

Proof.

For all n , $f(n) = (\Delta F)(n) = F(n+1) - F(n)$, so

$$\begin{aligned} f(0) + f(1) + \cdots + f(n) &= (F(1) - F(0)) + (F(2) - F(1)) \\ &\quad + \cdots + (F(n+1) - F(n)) \\ &= -F(0) + F(n+1) \\ &= F(n+1) - F(0). \end{aligned}$$



Example Using the FTDC

We derive the formula

$$1 + a + a^2 + \cdots + a^n = \frac{a^{n+1} - 1}{a - 1} \text{ if } a \neq 1.$$

Let $f(n) = a^n$, so $f(n) = (\Delta F)(n)$ for $F = ???$

Example Using the FTDC

We derive the formula

$$1 + a + a^2 + \cdots + a^n = \frac{a^{n+1} - 1}{a - 1} \text{ if } a \neq 1.$$

Let $f(n) = a^n$, so $f(n) = (\Delta F)(n)$ for $F = ???$

Recall for $f(n) = 3^n$ that $(\Delta f)(n) = 2 \cdot 3^n$. In general,

$$\begin{aligned} \Delta(a^n) = (a - 1)a^n &\implies \frac{1}{a - 1} \Delta(a^n) = a^n \text{ if } a \neq 1 \\ &\implies \Delta\left(\frac{1}{a - 1} a^n\right) = a^n. \end{aligned}$$

Example Using the FTDC

We derive the formula

$$1 + a + a^2 + \cdots + a^n = \frac{a^{n+1} - 1}{a - 1} \text{ if } a \neq 1.$$

Let $f(n) = a^n$, so $f(n) = (\Delta F)(n)$ for $F = ???$

Recall for $f(n) = 3^n$ that $(\Delta f)(n) = 2 \cdot 3^n$. In general,

$$\begin{aligned} \Delta(a^n) = (a - 1)a^n &\implies \frac{1}{a - 1} \Delta(a^n) = a^n \text{ if } a \neq 1 \\ &\implies \Delta\left(\frac{1}{a - 1} a^n\right) = a^n. \end{aligned}$$

Thus we can use $F(n) = \frac{a^n}{a - 1}$, so

$$\begin{aligned} 1 + a + a^2 + \cdots + a^n &= F(n + 1) - F(0) \\ &= \frac{a^{n+1}}{a - 1} - \frac{a^0}{a - 1} = \frac{a^{n+1} - 1}{a - 1}. \end{aligned}$$

Higher-Order Differences

We can repeatedly apply the difference operator to get the sequence $f, \Delta f, \Delta^2 f, \dots$: set $\Delta^0 f = f$, like $f^{(0)} = f$ in ordinary calculus, and

$$(\Delta f)(n) = f(n+1) - f(n)$$

$$(\Delta^2 f)(n) = (\Delta(\Delta f))(n) = (\Delta f)(n+1) - (\Delta f)(n)$$

$$\vdots$$

$$(\Delta^k f)(n) = (\Delta(\Delta^{k-1} f))(n) = (\Delta^{k-1} f)(n+1) - (\Delta^{k-1} f)(n)$$

for $k \geq 1$.

Higher-Order Differences

We can repeatedly apply the difference operator to get the sequence $f, \Delta f, \Delta^2 f, \dots$: set $\Delta^0 f = f$, like $f^{(0)} = f$ in ordinary calculus, and

$$(\Delta f)(n) = f(n+1) - f(n)$$

$$(\Delta^2 f)(n) = (\Delta(\Delta f))(n) = (\Delta f)(n+1) - (\Delta f)(n)$$

$$\vdots$$

$$(\Delta^k f)(n) = (\Delta(\Delta^{k-1} f))(n) = (\Delta^{k-1} f)(n+1) - (\Delta^{k-1} f)(n)$$

for $k \geq 1$.

Theorem

f is a polynomial sequence of degree d if and only if $\Delta^d f$ is a nonzero constant sequence. Moreover, $f(n) = c_d n^d + (\text{lower})$
 $\implies (\Delta^d f)(n) = d!c_d$ for all $n \geq 0$, where $c_d \neq 0$.

$\Delta^k f$ in terms of f

We can describe $\Delta^k f$ explicitly in terms of f :

$$(\Delta f)(n) = f(n+1) - f(n)$$

$$\begin{aligned}(\Delta^2 f)(n) &= (\Delta f)(n+1) - (\Delta f)(n) \\&= (f(n+2) - f(n+1)) - (f(n+1) - f(n)) \\&= f(n+2) - 2f(n+1) + f(n)\end{aligned}$$

$$\begin{aligned}(\Delta^3 f)(n) &= (\Delta^2 f)(n+1) - (\Delta^2 f)(n) \\&= f(n+3) - 3f(n+2) + 3f(n+1) - f(n)\end{aligned}$$

In general,

$$(\Delta^k f)(n) = \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} f(n+j)$$

Falling Powers

In ordinary calculus, $(x^k)' = kx^{k-1}$. But in discrete calculus, $\Delta(n^k) \neq kn^{k-1}$ for $k \geq 2$.

Definition

Set $n^{\underline{k}} = n(n-1) \cdots (n-(k-1))$ for $k \geq 1$ and $n^{\underline{0}} = 1$.

We call these **falling powers**. Note $n^{\underline{k}} = 0$ if $0 \leq n \leq k-1$.

Falling Powers

In ordinary calculus, $(x^k)' = kx^{k-1}$. But in discrete calculus, $\Delta(n^k) \neq kn^{k-1}$ for $k \geq 2$.

Definition

Set $n^{\underline{k}} = n(n-1) \cdots (n-(k-1))$ for $k \geq 1$ and $n^{\underline{0}} = 1$.

We call these **falling powers**. Note $n^{\underline{k}} = 0$ if $0 \leq n \leq k-1$.

Some Examples

$$n^{\underline{1}} = n$$

$$n = n^{\underline{1}}$$

$$n^{\underline{2}} = n^2 - n$$

$$n^2 = n^{\underline{2}} + n^{\underline{1}}$$

$$n^{\underline{3}} = n^3 - 3n^2 + 2n$$

$$n^3 = n^{\underline{3}} + 3n^{\underline{2}} + n^{\underline{1}}$$

$$n^{\underline{4}} = n^4 - 6n^3 + 11n^2 - 6n$$

$$n^4 = n^{\underline{4}} + 6n^{\underline{3}} + 7n^{\underline{2}} + n^{\underline{1}}$$

$$\vdots$$
$$\vdots$$

$$\text{Note: } \binom{n}{k} = \frac{n(n-1) \cdots (n-(k-1))}{k!} = \frac{n^{\underline{k}}}{k!} \text{ for } k \geq 0.$$

Example

Using $n^{\underline{k}} = n(n-1)\cdots(n-(k-1))$, let's compute $\Delta(n^{\underline{2}})$.

$$\begin{aligned}\Delta(n^{\underline{2}}) &= (n+1)^{\underline{2}} - n^{\underline{2}} \\ &= (n+1)n - n(n-1) \\ &= n^2 + n - n^2 + n \\ &= 2n = 2n^{\underline{1}}\end{aligned}$$

Example

Using $n^{\underline{k}} = n(n-1)\cdots(n-(k-1))$, let's compute $\Delta(n^{\underline{2}})$.

$$\begin{aligned}\Delta(n^{\underline{2}}) &= (n+1)^{\underline{2}} - n^{\underline{2}} \\ &= (n+1)n - n(n-1) \\ &= n^2 + n - n^2 + n \\ &= 2n = 2n^{\underline{1}}\end{aligned}$$

For $k \geq 0$, $\Delta(n^{\underline{k}}) = kn^{\underline{k-1}}$, so $n^{\underline{k}} = \Delta\left(\frac{n^{\underline{k+1}}}{k+1}\right)$. Thus $\{n^{\underline{k}}\}$ is a better sequence of “powers” than $\{n^k\}$ for discrete calculus.

Formulas for $1^k + 2^k + \cdots + n^k$ are complicated as k grows, but $1^{\underline{k}} + 2^{\underline{k}} + \cdots + n^{\underline{k}}$ is much nicer!

A falling power sum

For $k \neq -1$,

$$n^{\underline{k}} = \Delta \left(\frac{1}{k+1} n^{\underline{k+1}} \right)$$

$$\xrightarrow{\text{FTDC}} 1^{\underline{k}} + 2^{\underline{k}} + \cdots + n^{\underline{k}} = \sum_{j=0}^n j^{\underline{k}} = \frac{(n+1)^{\underline{k+1}}}{k+1}$$

since $0^{\underline{k+1}} = 0$, mirroring the behavior of $\int x^k dx = \frac{x^{k+1}}{k+1}$.

Converting Between Regular and Falling Powers

We saw earlier $n^3 = n^3 + 3n^2 + n^1$. How can we find the coefficients on the right in a direct way?

Converting Between Regular and Falling Powers

We saw earlier $n^3 = n^3 + 3n^2 + n^1$. How can we find the coefficients on the right in a direct way?

Since we can write $\{n^k\}$ and $\{n^{\underline{k}}\}$ in terms of each other, any polynomial sequence $f(n)$ has both forms:

$$f(n) = a_0 + a_1n + \cdots + a_dn^d = b_0 + b_1n^{\underline{d}} + \cdots + b_dn^{\underline{d}}$$

Example

$$1 + n + n^2 + n^3 = 1 + 3n^{\underline{1}} + 4n^{\underline{2}} + n^{\underline{3}}$$

Converting Between Regular and Falling Powers

How can we describe a_k and b_k in the equation

$$f(n) = a_0 + a_1n + \cdots + a_d n^d = b_0 + b_1 n^{\underline{d}} + \cdots + b_d n^{\underline{d}}$$

in terms of f ? We know $a_k = \frac{f^{(k)}(0)}{k!}$ from the Taylor expansion, and it turns out $b_k = \frac{(\Delta^k f)(0)}{k!}$.

Converting Between Regular and Falling Powers

How can we describe a_k and b_k in the equation

$$f(n) = a_0 + a_1n + \cdots + a_d n^d = b_0 + b_1 n^{\underline{1}} + \cdots + b_d n^{\underline{d}}$$

in terms of f ? We know $a_k = \frac{f^{(k)}(0)}{k!}$ from the Taylor expansion, and it turns out $b_k = \frac{(\Delta^k f)(0)}{k!}$.

Theorem

For a polynomial $f(n)$ of degree d ,

$$\begin{aligned} f(n) &= \sum_{k=0}^d \frac{(\Delta^k f)(0)}{k!} n^{\underline{k}} \\ &= \sum_{k=0}^d (\Delta^k f)(0) \frac{n(n-1)\cdots(n-(k-1))}{k!} \\ &= \sum_{k=0}^d (\Delta^k f)(0) \binom{n}{k}. \end{aligned}$$

Sum of First n Cubes

$$f(n) = n^3$$

$f(n)$	0	1	8	27	64	125
$(\Delta f)(n)$	1	7	19	37	61	...
$(\Delta^2 f)(n)$		6	12	18	24	...
$(\Delta^3 f)(n)$			6	6	6	...

Sum of First n Cubes

$$f(n) = n^3$$

$f(n)$	0	1	8	27	64	125
$(\Delta f)(n)$	1	7	19	37	61	...
$(\Delta^2 f)(n)$		6	12	18	24	...
$(\Delta^3 f)(n)$			6	6	6	...

We see that

k	0	1	2	3	4	5	...
$(\Delta^k f)(0)$	0	1	6	6	0	0	...

so

$$\begin{aligned} n^3 &= \sum_{k=0}^3 (\Delta^k f)(0) \binom{n}{k} = n + 6 \binom{n}{2} + 6 \binom{n}{3} \\ &= n^1 + 3n^2 + n^3. \end{aligned}$$

Sum of First n Cubes

Recalling that $n^3 = n^3 + 3n^2 + n^1$ lets us derive the formula for summing the first n cubes:

$$\begin{aligned}\sum_{j=0}^n j^3 &= \sum_{j=0}^n [j^3 + 3j^2 + j^1] \\&= \frac{(n+1)^4}{4} + \frac{3(n+1)^3}{3} + \frac{(n+1)^2}{2} \\&= \frac{(n+1)n(n-1)(n-2)}{4} + (n+1)n(n-1) + \frac{(n+1)n}{2} \\&= \frac{n^4}{4} + \frac{n^3}{2} + \frac{n^2}{4} \\&= \frac{n^4 + 2n^3 + n^2}{4} \\&= \frac{n^2(n+1)^2}{4}\end{aligned}$$

Thank you!

Example Using Higher-Order Differences

$$f(n) = n^3$$

$f(n)$	0	1	8	27	64	125
$(\Delta f)(n)$	1	7	19	37	61	...
$(\Delta^2 f)(n)$		6	12	18	24	...
$(\Delta^3 f)(n)$			6	6	6	...
$(\Delta^4 f)(n)$			0	0	...	

Example Using Higher-Order Differences

$f(n) = n^3$							
$f(n)$	0	1	8	27	64	125	
$(\Delta f)(n)$		1	7	19	37	61	...
$(\Delta^2 f)(n)$			6	12	18	24	...
$(\Delta^3 f)(n)$			6	6	6	...	
$(\Delta^4 f)(n)$				0	0	...	

We have

$$f(n) = n^3$$

$$(\Delta f)(n) = 3n^2 + 3n + 1$$

$$(\Delta^2 f)(n) = 6n + 6$$

$$(\Delta^3 f)(n) = 6$$

$$(\Delta^k f)(n) = 0 \text{ for } k \geq 4.$$

Discrete Derivative

How can we define $\Delta(n^k)$ for $k < 0$ to make $\Delta(n^k) = kn^{k-1}$ when $k < 0$?

Definition

When $k > 0$, set

$$n^{-k} = \frac{1}{(n+1)(n+2)\cdots(n+k)}.$$

Then $\Delta(n^k) = kn^{k-1}$ for all $k \in \mathbb{Z}$.

For example, $n^{-1} = \frac{1}{n+1}$ and $n^{-2} = \frac{1}{(n+1)(n+2)}$.

Polynomial Interpolation

Suppose we want to find a sequence such that $f(0) = 1$, $f(1) = 2$, $f(2) = 3$, and $f(3) = C$. We can find $f(x)$ that is a polynomial of degree at most 3.

Polynomial Interpolation

Suppose we want to find a sequence such that $f(0) = 1$, $f(1) = 2$, $f(2) = 3$, and $f(3) = C$. We can find $f(x)$ that is a polynomial of degree at most 3.

Standard approach:

Write $f(x) = a_3x^3 + a_2x^2 + a_1x + a_0$ with unknown coefficients and use the four conditions to get a system of equations:

$$f(0) = 1 \implies a_0 = 1$$

$$f(1) = 2 \implies a_3 + a_2 + a_1 + a_0 = 2$$

$$f(2) = 3 \implies 8a_3 + 4a_2 + 2a_1 + a_0 = 3$$

$$f(3) = C \implies 27a_3 + 9a_2 + 3a_1 + a_0 = C$$

We could instead work with a linear combination of $\binom{n}{k}$ rather than n^k and the coefficients come from finite differences:

Polynomial Interpolation

We could instead work with a linear combination of $\binom{n}{k}$ rather than n^k and the coefficients come from finite differences:

$f(n)$	1	2	3	C
$(\Delta f)(n)$	1	1		$C - 3$
$(\Delta^2 f)(n)$		0	$C - 4$	
$(\Delta^3 f)(n)$			$C - 4$	

We have $f(0) = 1$, $(\Delta f)(0) = 1$, $(\Delta^2 f)(0) = 0$, and $(\Delta^3 f)(0) = C - 4$.

Polynomial Interpolation

Taking $(\Delta^k f)(0) = 0$ for all $k \geq 4$, we get

$$\begin{aligned} f(n) &= \sum_{k=0}^3 (\Delta^k f)(0) \binom{n}{k} \\ &= 1 + \binom{n}{1} + 0 \binom{n}{2} + (C - 4) \binom{n}{3} \\ &= (C - 4) \binom{n}{3} + n + 1 \\ &= \frac{(C - 4)n(n - 1)(n - 2)}{6} + n + 1. \end{aligned}$$

$f(n)$ is cubic in n and satisfies our initial conditions $f(0) = 1$, $f(1) = 2$, $f(2) = 3$, and $f(3) = C$.

Application to Power Series

Let's use $n^3 = n^3 + 3n^2 + n^1$ to derive a formula for $\sum_{n \geq 1} n^3 x^n$

when $|x| < 1$ systematically.

Start with $\sum_{n \geq 1} x^n = \frac{x}{1-x}$ and differentiate three times:

Application to Power Series

Let's use $n^3 = n^3 + 3n^2 + n^1$ to derive a formula for $\sum_{n \geq 1} n^3 x^n$

when $|x| < 1$ systematically.

Start with $\sum_{n \geq 1} x^n = \frac{x}{1-x}$ and differentiate three times:

$$\sum_{n \geq 1} nx^{n-1} = \frac{1}{(1-x)^2} \implies \sum_{n \geq 1} n^1 x^n = \frac{x}{(1-x)^2}$$

$$\sum_{n \geq 1} n(n-1)x^{n-2} = \frac{2}{(1-x)^3} \implies \sum_{n \geq 1} n^2 x^n = \frac{2x^2}{(1-x)^3}$$

$$\sum_{n \geq 1} n(n-1)(n-2)x^{n-3} = \frac{6}{(1-x)^4} \implies \sum_{n \geq 1} n^3 x^n = \frac{6x^3}{(1-x)^4}$$

Application to Power Series

In general, when $|x| < 1$ we have

$$\sum_{n \geq 1} n^k x^{n-k} = \frac{k!}{(1-x)^{k+1}} \implies \sum_{n \geq 1} n^k x^n = \frac{k! x^k}{(1-x)^{k+1}}.$$

Application to Power Series

In general, when $|x| < 1$ we have

$$\sum_{n \geq 1} n^k x^{n-k} = \frac{k!}{(1-x)^{k+1}} \implies \sum_{n \geq 1} n^k x^n = \frac{k! x^k}{(1-x)^{k+1}}.$$

Since $n^3 = n^3 + 3n^2 + n^1$,

$$\begin{aligned} \sum_{n \geq 1} n^3 x^n &= \sum_{n \geq 1} (n^3 + 3n^2 + n^1) x^n \\ &= \frac{6x^3}{(1-x)^4} + 3 \frac{2x^2}{(1-x)^3} + \frac{x}{(1-x)^2} \\ &= \frac{x^3 + 4x^2 + x}{(1-x)^4}. \end{aligned}$$